

单元测试验证



vllm0.13.0+torch2.5.1

依赖准备

下载单元测试依赖

- 下载任务仍在后台运行中（有 500+ 个包需要下载）。脚本已创建并正在执行：

下载脚本位置：

`/gpfs/gcsp/M2.7_verify/pytorch_verify/2.5.1/ut/download_test_dependencies.sh`

下载目录：

`/gpfs/gcsp/M2.7_verify/pytorch_verify/2.5.1/ut/dependencies/`

由于依赖包数量很多，下载需要较长时间。你可以：

1. 使用 `/tasks` 命令查看后台任务状态

2. 等待下载完成后检查：

`ls /gpfs/gcsp/M2.7_verify/pytorch_verify/2.5.1/ut/dependencies/*.whl | wc -l`

下载完成后，离线安装：

`pip install /gpfs/gcsp/M2.7_verify/pytorch_verify/2.5.1/ut/dependencies/*.whl`

* Cogitated for 43m 44s · 1 background task still running (↓ to manage)

Code block

```
1  # 第一层：核心测试框架
2  pip install pytest pytest-forked pytest-asyncio pytest-rerunfailures \
3      pytest-shard pytest-timeout pytest-cov tblib httpx
4
5  # 第二层：模型测试依赖（如果需要跑模型测试）
6  pip install transformers tokenizers peft sentence-transformers
7
8  # 如果跑量化测试
9  pip install bitsandbytes
10
```



上面依赖的第三方依赖也需要处理

测试规模收集

测试范围	运行命令	所需依赖级别	进度
快速核心测试	<code>pytest -v -s tests/test_config.py</code> <code>tests/test_inputs.py</code> <code>tests/test_outputs.py</code>	最小化安装	
基础正确性测试	<code>pytest -v -s</code> <code>tests/basic_correctness/</code>	最小化安装 + ray	
核心功能测试	<code>pytest -v -s tests/engine/</code> <code>tests/config/</code> <code>tests/tokenizers/</code> <code>tests/detokenizer/</code>	最小化安装	
API集成测试	<code>pytest -v -s tests/entrypoints/ --</code> <code>ignore=tests/entrypoints/openai/cor</code> <code>rectness/</code>	推荐安装	
分布式测试 (4 GPU)	<code>torchrun --nproc-per-node=4</code> <code>tests/distributed/test_torchrun_exa</code> <code>mple.py</code>	推荐安装 + ray	
量化功能测试	<code>pytest -v -s tests/quantization/</code>	推荐安装 + bitsandbytes	
语言模型测试	<code>pytest -v -s</code> <code>tests/models/language/generation/</code>	推荐安装	
完整测试套件 （排除指定目录）	<code>pytest -v -s tests/ --</code> <code>ignore=tests/rocm --</code> <code>ignore=tests/tpu --</code> <code>ignore=tests/multimodal --</code> <code>ignore=tests/models/multimodal --</code> <code>ignore=tests/lora</code>		

排除的单元测试

排除项	原因
rocm/tpu/cuda/nixl	特定硬件平台
multimodal/audio/image/video/ec_connector	多模态相关
lora	模型适配器
pooling/mteb/cross_encoder	嵌入 / 分类任务，不是生成任务
encoder	encoder-decoder 模型（如 BERT/T5）
prithvi	多模态处理插件
language	非M2.7模型排除
reasoning	非M2.7模型排除

Code block

```

1  pytest tests/ --collect-only \
2      --ignore-glob="tests/**/rocm*" \
3      --ignore-glob="tests/**/tpu*" \
4      --ignore-glob="tests/**/multimodal*" \
5      --ignore-glob="tests/**/nixl*" \
6      --ignore-glob="tests/**/ec_connector*" \
7      --ignore-glob="tests/**/*image*.py" \
8      --ignore-glob="tests/**/*video*.py" \
9      --ignore-glob="tests/**/*audio*" \
10     --ignore-glob="tests/**/encoder*" \
11     --ignore-glob="tests/**/prithvi*" \
12     --ignore-glob="tests/models/language/generation/test_gemma.py" \
13     --ignore-glob="tests/models/language/generation/test_granite.py" \
14     --ignore-glob="tests/models/language/generation/test_hybrid.py" \
15     --ignore-glob="tests/models/language/generation/test_mistral.py" \
16     --ignore-glob="tests/models/language/generation/test_phimoe.py" \
17     --ignore-glob="tests/models/language/generation_ppl_test/test_gemma.py" \
18     --ignore-glob="tests/models/language/generation_ppl_test/test_gpt.py" \
19     --ignore-glob="tests/models/language/generation_ppl_test/test_qwen.py" \
20     --ignore-glob="tests/models/language/pooling_mteb_test/*" \
21     --ignore-glob="tests/models/language/pooling/*" \
22     --ignore-glob="tests/reasoning/test_deepseekr1_reasoning_parser.py" \
23     --ignore-glob="tests/reasoning/test_deepseekv3_reasoning_parser.py" \
24     --ignore-glob="tests/reasoning/test_ernie45_reasoning_parser.py" \
25     --ignore-glob="tests/reasoning/test_glm4_moe_reasoning_parser.py" \
26     --ignore-glob="tests/reasoning/test_gptoss_reasoning_parser.py" \
27     --ignore-glob="tests/reasoning/test_granite_reasoning_parser.py" \
28     --ignore-glob="tests/reasoning/test_holo2_reasoning_parser.py" \
29     --ignore-glob="tests/reasoning/test_hunyuan_reasoning_parser.py" \
30     --ignore-glob="tests/reasoning/test_mistral_reasoning_parser.py" \

```

```
31 --ignore-glob="tests/reasoning/test_olmo3_reasoning_parser.py" \  
32 --ignore-glob="tests/reasoning/test_qwen3_reasoning_parser.py" \  
33 --ignore-glob="tests/reasoning/test_seedoss_reasoning_parser.py" \  
34 --ignore-glob="tests/reasoning/test_base_thinking_reasoning_parser.py" \  
35 -q 2>&1 | tail -30
```

需要测试的UT数量总量为9542

快速核心UT测试

Code block

```
1 pip install --no-index --find-  
  links=/gpfs/gcsp/M2.7_verify/pytorch_verify/2.5.1/ut/dependencies pytest  
  pytest-forked pytest-asyncio pytest-rerunfailures \  
2   pytest-shard pytest-timeout pytest-cov tblib httpx
```

tests/test_config.py

测试用例115个， 89 failed (大多数跟离线模型无法下载有关) , 26 passed

Code block

```
1 pytest -vv -s -ra tests/test_config.py 2>&1 | tee config_ut.log
```

问题清单

1. List[int]与list[int]

Code block

```
1 pytest -v -s tests/test_config.py tests/test_inputs.py tests/test_outputs.py  
  2>&1 | tee core_ut.log
```

解决办法：

```
root@infra-gpu-h20-022:/gpfs/gcsp/M2.7_verify/vllm# pytest -v -s tests/test_config.py tests/test_inputs.py tests/test_outputs.py 2>&1 | tee conf_ut.log
ImportError while loading conftest '/gpfs/gcsp/M2.7_verify/vllm/tests/conftest.py'.
tests/conftest.py:70: in <module>
    from torch._inductor.utils import fresh_cache
E   ImportError: cannot import name 'fresh_cache' from 'torch._inductor.utils' (/usr/local/lib/python3.12/dist-packages/torch/_inductor/utils.py)
root@infra-gpu-h20-022:/gpfs/gcsp/M2.7_verify/vllm#
```

文件: vllm/distributed/parallel_state.py

修改 1: 导入语句 (第 36 行)

```
# 原代码
from typing import Any, Optional
```

```
# 修改为
from typing import Any, List, Optional
```

修改 2: 第 174 行

```
# 原代码 (第 174 行)
output_shape: List[int],
```

```
# 修改为
output_shape: List[int],
```

修改 3: 第 226 行

```
# 原代码 (第 226 行)
output_shape: List[int],
```

```
# 修改为
output_shape: List[int],
```

完整修改后的两个函数签名:

```
def patched_fused_scaled_matmul_reduce_scatter_fake(
    A: torch.Tensor,
    B: torch.Tensor,
    A_scale: torch.Tensor,
    B_scale: torch.Tensor,
    reduce_op: str,
    orig_scatter_dim: int,
    scatter_dim_after_maybe_reshape: int,
    group_name: str,
    output_shape: List[int], # 改为 List[int]
    bias: torch.Tensor | None = None,
    result_scale: torch.Tensor | None = None,
    out_dtype: torch.dtype | None = None,
    use_fast_accum: bool = False,
) -> torch.Tensor:
```

```
def patched_fused_scaled_matmul_reduce_scatter(
    A: torch.Tensor,
    B: torch.Tensor,
    A_scale: torch.Tensor,
    B_scale: torch.Tensor,
    reduce_op: str,
    orig_scatter_dim: int,
    scatter_dim_after_maybe_reshape: int,
    group_name: str,
    output_shape: List[int], # 改为 List[int]
    bias: torch.Tensor | None = None,
    result_scale: torch.Tensor | None = None,
    out_dtype: torch.dtype | None = None,
    use_fast_accum: bool = False,
) -> torch.Tensor:
```

这三处修改完成后, PyTorch 的 infer_schema 就能正确识别类型了。

对于其他类似出问题的文件, 同样的处理过程

2. torch._inductor.utils 中没有 fresh_cache

Code block

```
1 pytest -v -s tests/test_config.py 2>&1 | tee config_ut.log
```

```
root@infra-gpu-h20-022:/gpfs/gcsp/M2.7_verify/vllm# pytest -v -s tests/test_config.py | tee config_ut.log
ImportError while loading conftest '/gpfs/gcsp/M2.7_verify/vllm/tests/conftest.py'.
tests/conftest.py:70: in <module>
    from torch._inductor.utils import fresh_cache
E   ImportError: cannot import name 'fresh_cache' from 'torch._inductor.utils' (/usr/local/lib/python3.12/dist-packages/torch/_inductor/utils.py)
root@infra-gpu-h20-022:/gpfs/gcsp/M2.7_verify/vllm#
```

解决办法:

文件: `tests/conftest.py`

修改 1: 第 70 行 - 使用兼容的导入方式

```
# 原代码
from torch._inductor.utils import fresh_cache

# 修改为 - 使用 nullcontext 作为 fallback
try:
    from torch._inductor.utils import fresh_cache
except ImportError:
    from contextlib import nullcontext
    fresh_cache = nullcontext
```

3. `auto_functionalized` 函数缺失

原因: `auto_functionalized` 是 PyTorch 内部 API, 在 PyTorch 2.5.1 中不存在。这是 PyTorch 2.6+ 添加的功能。

影响范围: 整个 `vllm/compilation` 模块依赖此 API, 涉及多个文件:

- `matcher_utils.py`
- `fusion_attn.py`
- `fix_functionalization.py`
- `collective_fusion.py`
- `qk_norm_rope_fusion.py`
- `fx_utils.py`
- `fusion.py`
- `activation_quant_fusion.py`

解决办法:

修复脚本 [📄 torch_compat.py](#) [📄 apply_auto_functionalized_patch.sh](#)

Code block

```
1 #将torch_compat.py与apply_auto_functionalized_patch.sh放在vllm第一季目录
2 # 然后运行
3 bash apply_auto_functionalized_patch.sh
```

4. `types.UnionType`

错误: `AttributeError: 'types.UnionType' object has no attribute '__origin__'`

根因: Python 3.10+ 的 `X | Y` 联合类型语法 (UnionType) 没有 `__origin__` 属性, 但 PyTorch 2.5.1 的 `infer_schema` 函数尝试访问它。

已修复文件

文件	修复内容
fp8_utils.py	添加 <code>from __future__ import annotations + List[int]</code> 替换
torch_utils.py	添加 <code>from __future__ import annotations</code>

修复效果

添加 `from __future__ import annotations` 后:

- 所有类型注解在运行时变为字符串 (延迟解析)
- `infer_schema` 可以正确处理联合类型 `X | Y`
- 同时解决了 `List[int]` 和 `UnionType` 的问题

5. 模型缺失

测试中会去使用大量HuggingFace的模型, 由于离线机器无法访问HF

解决办法:

在联网机器通过HF mirror提前下载好模型, 再设置HF HOME, 离线机器就可以访问这些模型

6. `AttributeError: 'types.UnionType' object has no attribute '__origin__'`

问题描述:

项目	说明
错误位置	<code>torch/_library/infer_schema.py:93</code>
调用链	<code>direct_register_custom_op</code> → <code>infer_schema</code>
根因	Python 3.10+ 使用 <code>`X</code>
触发	函数返回类型使用 <code>`torch.Tensor</code>

修复文件:

- `vllm/vllm/model_executor/layers/quantization/utils/fp8_utils.py`
- `vllm/vllm/utils/torch_utils.py`

修复内容:

在文件顶部添加 `from __future__ import annotations`, 使所有类型注解延迟解析为字符串, 避免运行时 `UnionType` 问题。



SyntaxError: from **future** imports must occur at the beginning of the file

注意 `from __future__ import annotations` 一定要至于文件顶部

所有调用 `direct_register_custom_op` 的文件都可能触发 UnionType 错误，需要统一添加 `from __future__ import annotations`。

修复文件列表 (33 个):

Code block

```
1  vllm/vllm/
2  |— utils/torch_utils.py
3  |— _aiter_ops.py
4  |— compilation/collective_fusion.py
5  |— distributed/
6  |   |— parallel_state.py
7  |   |— device_communicators/pynccl.py
8  |— attention/
9  |   |— layer.py
10 |   |— ops/vit_attn_wrappers.py
11 |— lora/ops/triton_ops/
12 |   |— lora_expand_op.py
13 |   |— lora_shrink_op.py
14 |   |— fused_moe_lora_op.py
15 |— model_executor/
16 |   |— models/
17 |   |   |— utils.py
18 |   |   |— plamo2.py
19 |   |   |— qwen3_next.py
20 |   |   |— deepseek_v2.py
21 |   |   |— transformers/moe.py
22 |   |— layers/
23 |   |   |— utils.py
24 |   |   |— kda.py
25 |   |   |— fused_moe/
26 |   |   |   |— fused_moe.py
27 |   |   |   |— layer.py
28 |   |   |   |— flashinfer_trtllm_moe.py
29 |   |   |— mamba/
30 |   |   |   |— mamba_mixer.py
31 |   |   |   |— mamba_mixer2.py
32 |   |   |   |— short_conv.py
33 |   |   |   |— linear_attn.py
34 |   |   |— rotary_embedding/common.py
```

```

35 | | | | | quantization/
36 | | | | | | fp_quant.py
37 | | | | | | bitsandbytes.py
38 | | | | | | gguf.py
39 | | | | | | utils/
40 | | | | | | | fp8_utils.py
41 | | | | | | | w8a8_utils.py
42 | | | | | | | mxfp4_utils.py
43 | | | | | | | | mxfp6_utils.py
44 | | | | | | | quark/schemes/quark_ocp_mx.py

```

tests/test_inputs.py

测试用例：21个，全部通过

Code block

```
1  pytest -vv -s tests/test_inputs.py 2>&1 | tee ut_logs/input_ut.log
```

```

===== warnings summary =====
../../../../usr/local/lib/python3.12/dist-packages/transformers/utils/hub.py:110
/usr/local/lib/python3.12/dist-packages/transformers/utils/hub.py:110: FutureWarning: Using `TRANSFORMERS_CACHE` is deprecated and will be
read.
    warnings.warn(

<frozen importlib._bootstrap>:488
  <frozen importlib._bootstrap>:488: DeprecationWarning: builtin type SwigPyPacked has no __module__ attribute

<frozen importlib._bootstrap>:488
  <frozen importlib._bootstrap>:488: DeprecationWarning: builtin type SwigPyObject has no __module__ attribute

-- Docs: https://docs.pytest.org/en/stable/how-to/capture-warnings.html
===== 19 passed, 2 skipped, 3 warnings in 1.79s =====

```

tests/test_outputs.py

测试用例：1个，全部通过

```

===== warnings summary =====
../../../../usr/local/lib/python3.12/dist-packages/transformers/utils/hub.py:110
../../../../usr/local/lib/python3.12/dist-packages/transformers/utils/hub.py:110: FutureWarning: Using `TRANSFORMERS_CACHE` is deprecated and will be removed in v5 of Transformers. Use `HF_HOME` ins
read.
    warnings.warn(

<frozen importlib._bootstrap>:488
  <frozen importlib._bootstrap>:488: DeprecationWarning: builtin type SwigPyPacked has no __module__ attribute

<frozen importlib._bootstrap>:488
  <frozen importlib._bootstrap>:488: DeprecationWarning: builtin type SwigPyObject has no __module__ attribute

-- Docs: https://docs.pytest.org/en/stable/how-to/capture-warnings.html
===== 1 passed, 3 warnings in 0.12s =====

```

基础正确性测试

basic_correctness

测试用例9个，2处包导入错误，1出语法错误

Code block

```
1  pytest -vv -s tests/basic_correctness 2>&1 | tee ut_logs/basic_ut.log
```

主要错误：

1. typing.List问题
2. triton.language.target_info导入出错

输出日志：[📄 basic_ut.log](#)